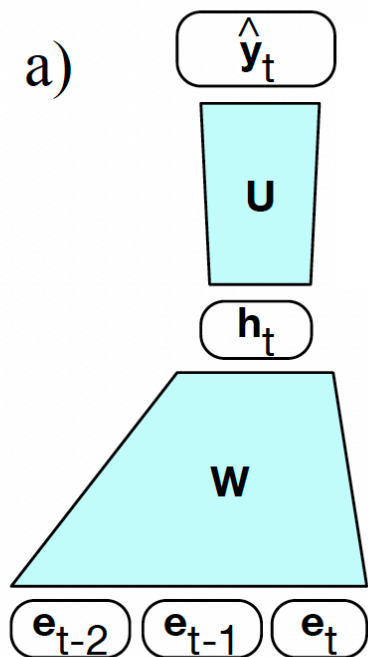
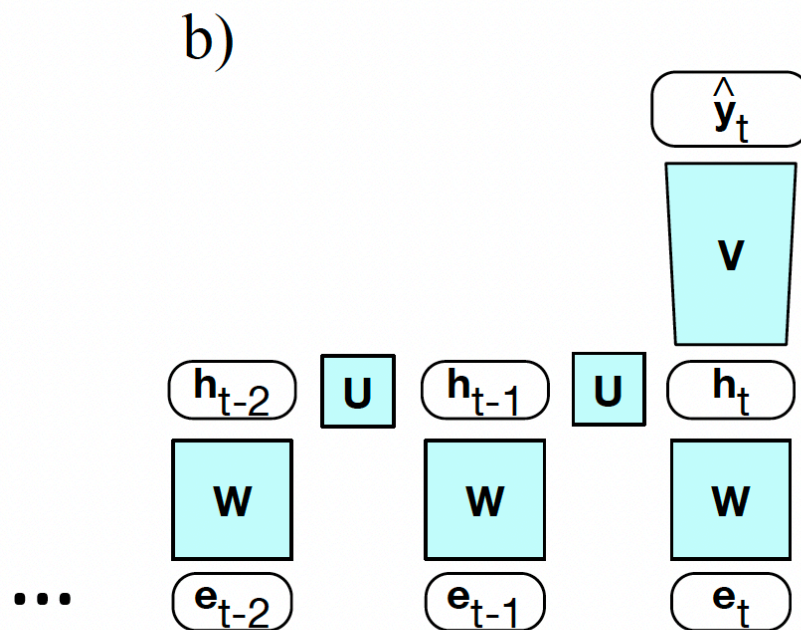

Neural Networks For NLP

FFN LMs vs RNN LMs



FNN

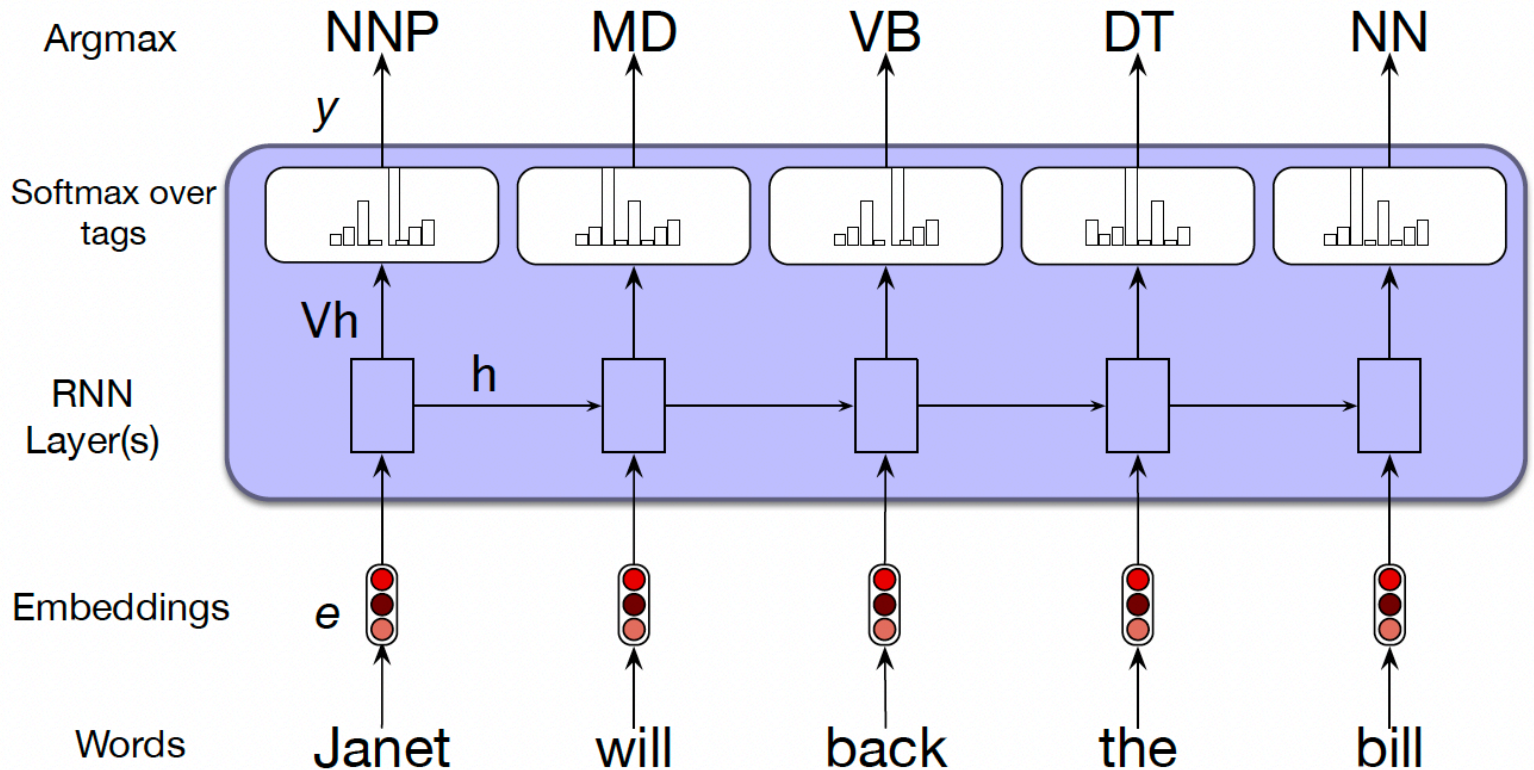


RNN

RNNs for sequence labeling

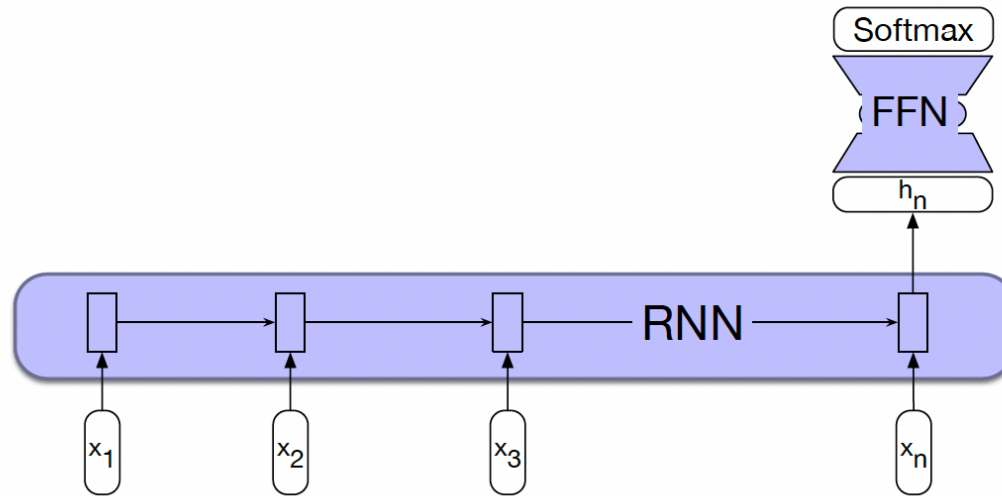
Assign a label to each element of a sequence

Part-of-speech tagging



RNNs for sequence classification

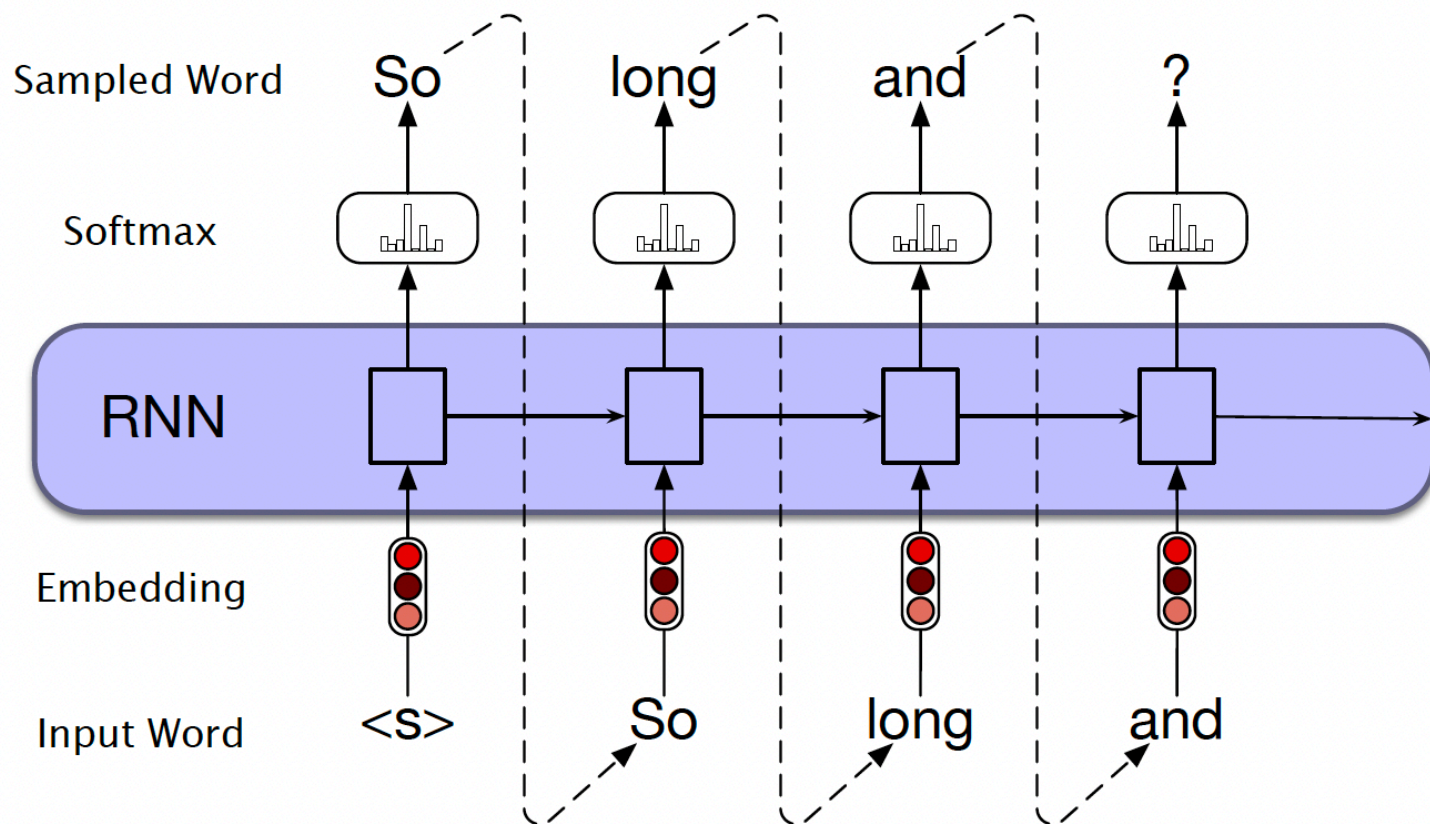
Text classification



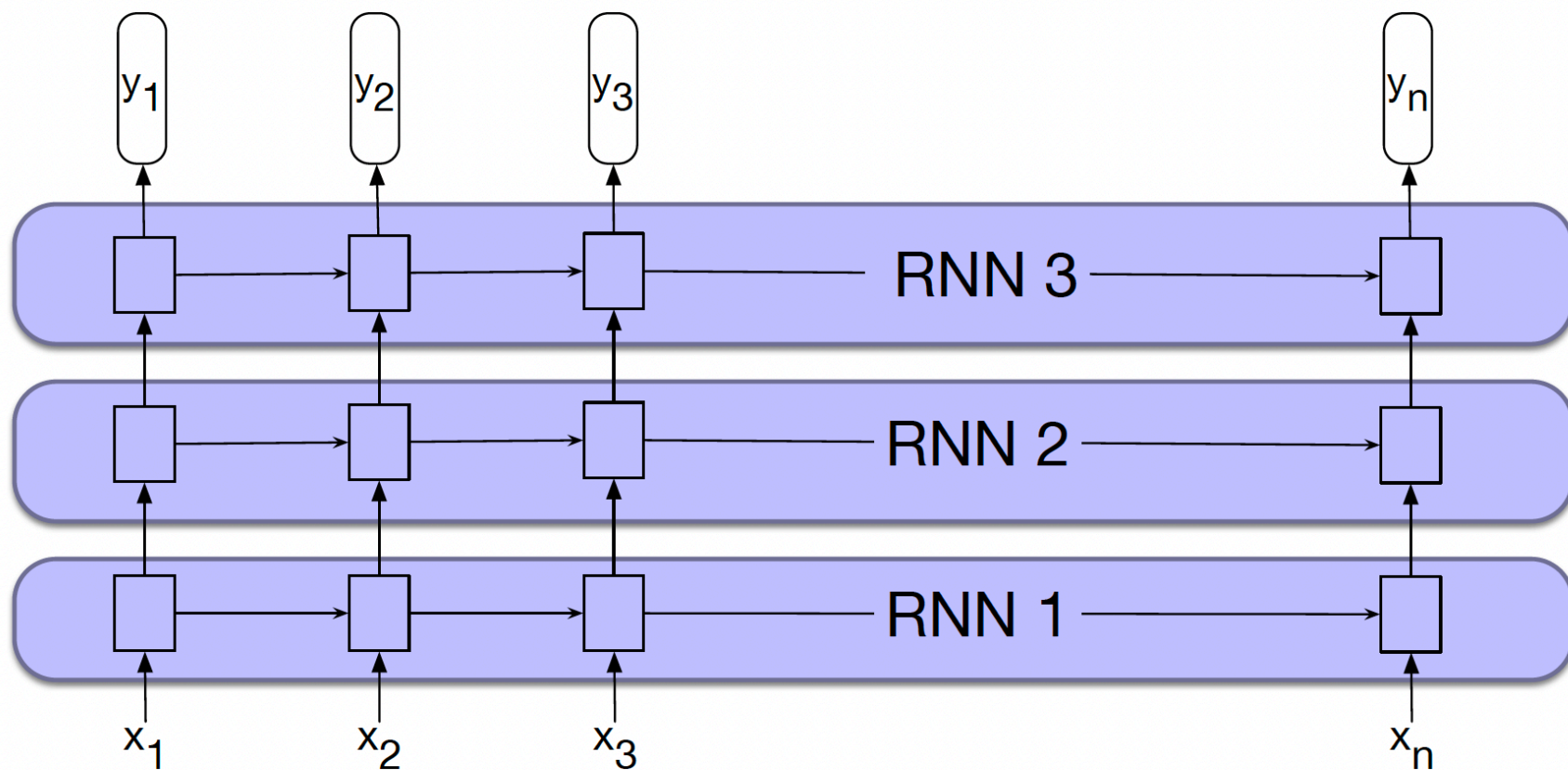
Instead of taking the last state, could use some pooling function of all the output states, like **mean pooling**

$$\mathbf{h}_{mean} = \frac{1}{n} \sum_{i=1}^n \mathbf{h}_i$$

Autoregressive generation



Stacked RNNs

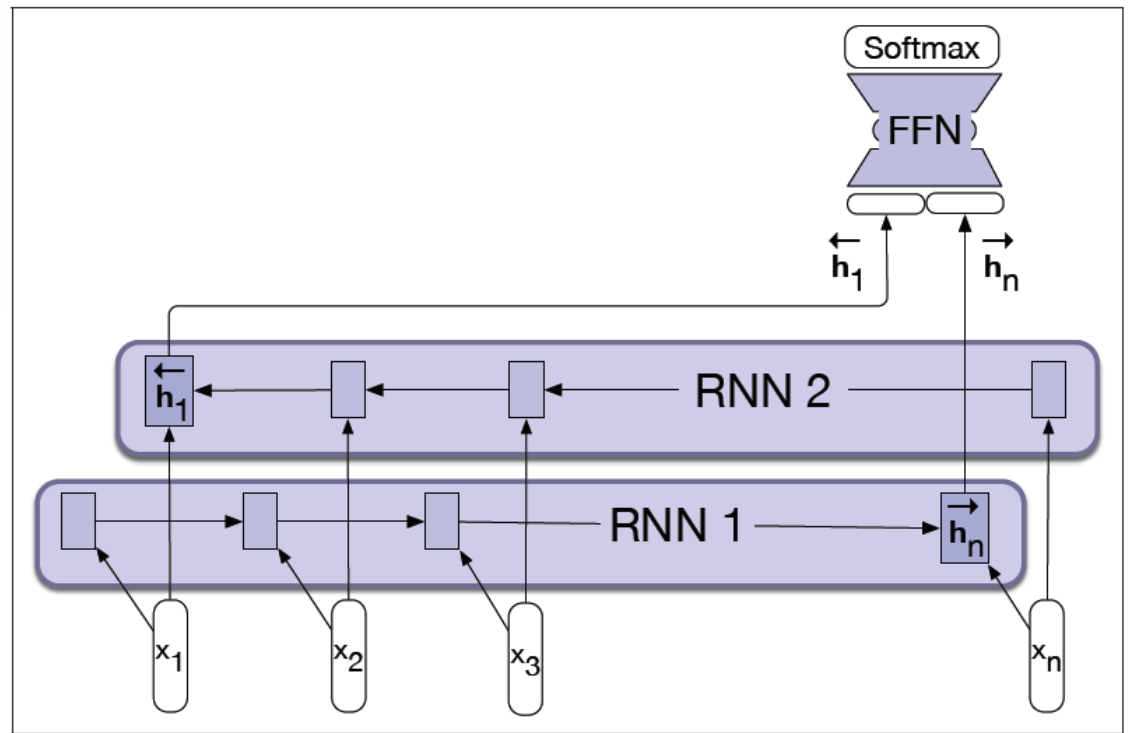


Bidirectional RNNs

$$\mathbf{h}_t^f = \text{RNN}_{\text{forward}}(\mathbf{x}_1, \dots, \mathbf{x}_t)$$

$$\mathbf{h}_t^b = \text{RNN}_{\text{backward}}(\mathbf{x}_t, \dots, \mathbf{x}_n)$$

$$\begin{aligned}\mathbf{h}_t &= [\mathbf{h}_t^f; \mathbf{h}_t^b] \\ &= \mathbf{h}_t^f \oplus \mathbf{h}_t^b\end{aligned}$$



The flights the airline was cancelling were full

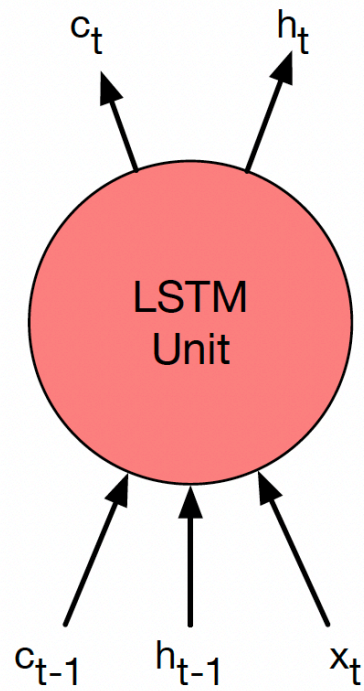
Motivating the LSTM: dealing with distance

- It's hard to assign probabilities accurately when context is very far away
 - The flights the airline was canceling were full.
- Hidden layers are being forced to do two things:
 - Provide information useful for the current decision,
 - Update and carry forward information required for future decisions.
- Another problem: During backprop, we have to repeatedly multiply gradients through time and many h 's
 - The "vanishing gradient" problem

The LSTM: Long short-term memory network

LSTMs divide the context management problem into two subproblems:

- removing information no longer needed from the context,
- adding information likely to be needed for later decision making
- LSTMs add:
 - explicit context layer
 - Neural circuits with **gates** to control information flow



LSTM

Forget gate

Deletes information from the context that is no longer needed.

$$\mathbf{f}_t = \sigma(\mathbf{U}_f \mathbf{h}_{t-1} + \mathbf{W}_f \mathbf{x}_t)$$

$$\mathbf{k}_t = \mathbf{c}_{t-1} \odot \mathbf{f}_t$$

Add gate

$$\mathbf{g}_t = \tanh(\mathbf{U}_g \mathbf{h}_{t-1} + \mathbf{W}_g \mathbf{x}_t)$$

Selecting information to add to current context

$$\mathbf{i}_t = \sigma(\mathbf{U}_i \mathbf{h}_{t-1} + \mathbf{W}_i \mathbf{x}_t)$$

$$\mathbf{j}_t = \mathbf{g}_t \odot \mathbf{i}_t$$

Add this to the modified context vector to get our new context vector.

$$\mathbf{c}_t = \mathbf{j}_t + \mathbf{k}_t$$

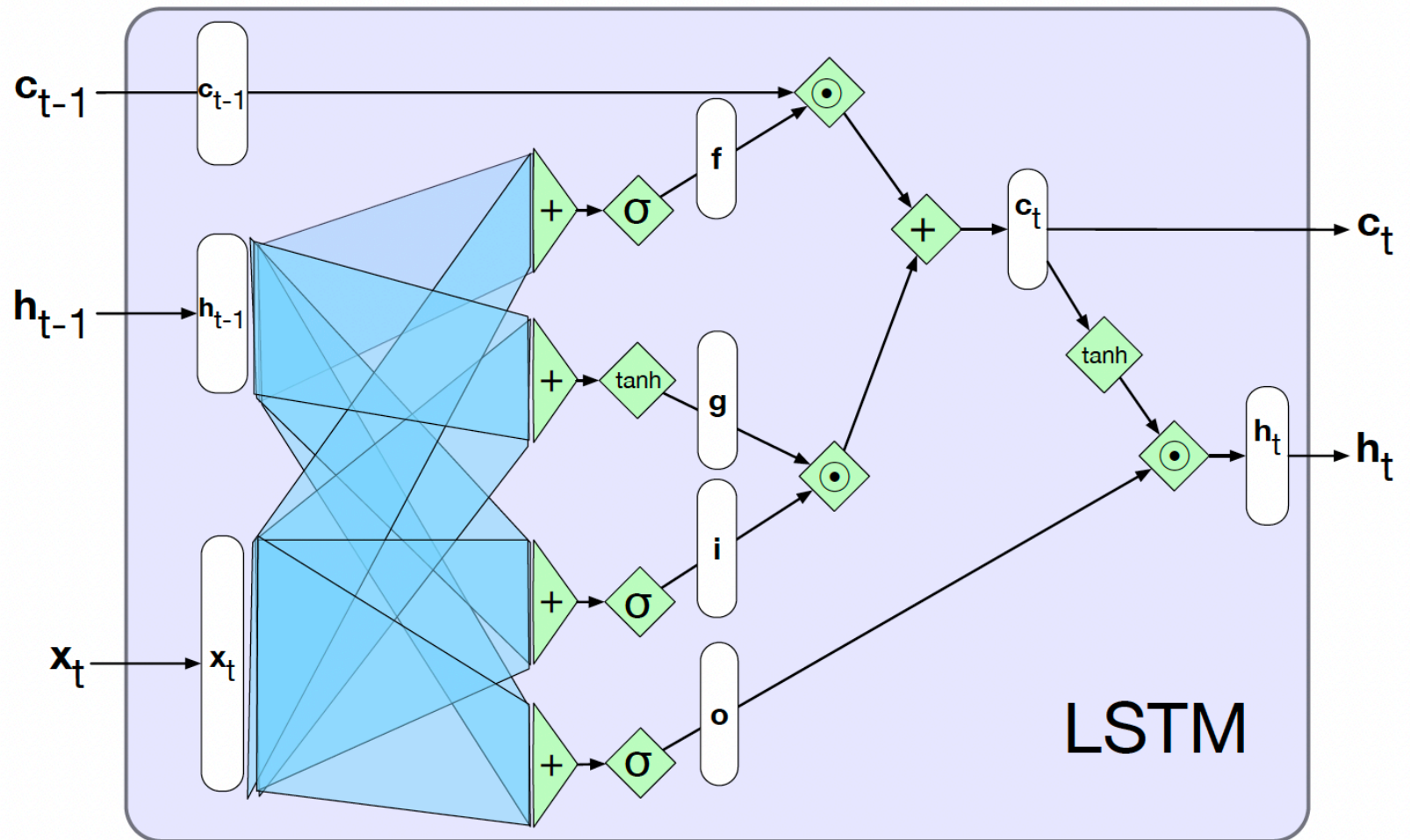
Output gate

Decide what information is required for the current hidden state (as opposed to what information needs to be preserved for future decisions).

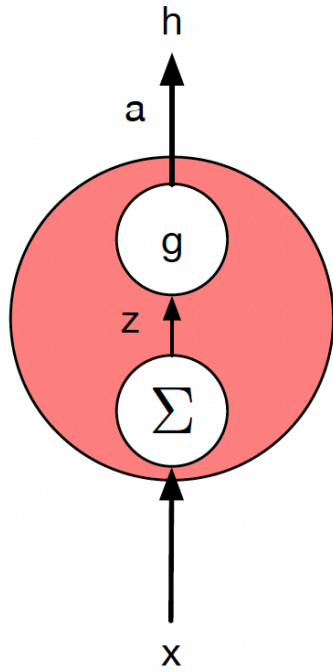
$$\mathbf{o}_t = \sigma(\mathbf{U}_o \mathbf{h}_{t-1} + \mathbf{W}_o \mathbf{x}_t)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t)$$

The LSTM

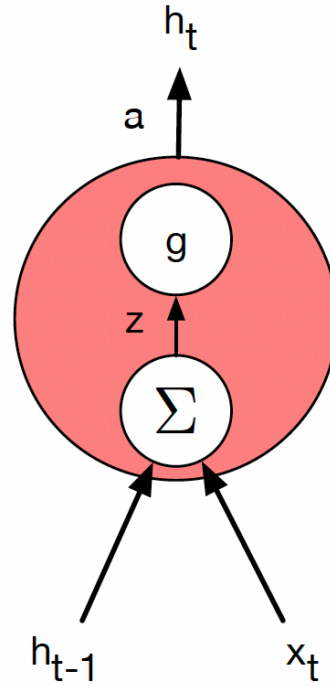


Units



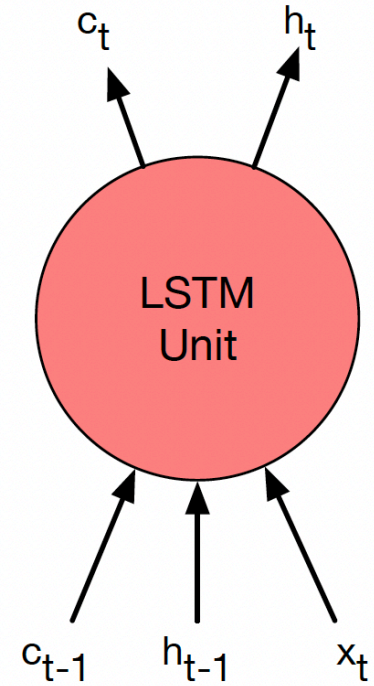
(a)

FNN



(b)

RNN



(c)

LSTM